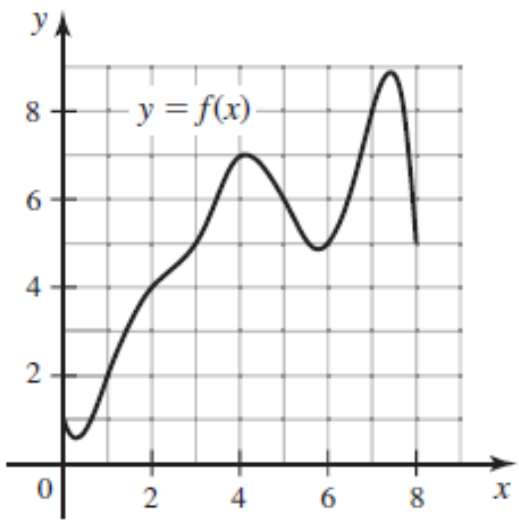
# Problem Set 2C

**Name:** _____

**Instructions:** Show all work in a neat and organized manner. Unsupported answers will not receive full credit. If a problem requires multiple steps, clearly label and justify each step.

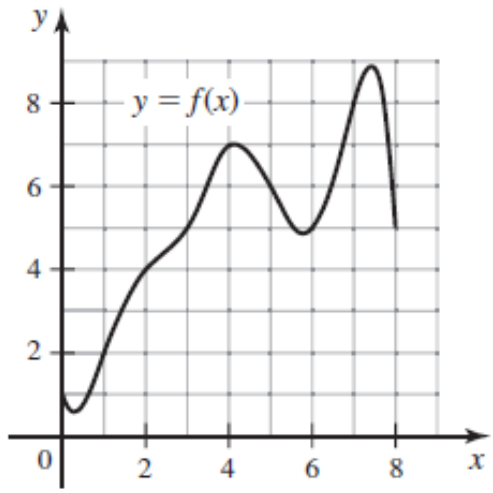
1. Using the given graph of $y = f(x)$




(a) Compute the Midpoint Rule approximation $M(4)$ for $\int_0^8 f(x)\,dx$. Be sure to write out the sum you used to find your approximation.

(b) Using the given graph of $y = f(x)$, compute the Trapezoid Rule approximation $T(4)$ for $\int_0^8 f(x)\,dx$.Be sure to write out the sum you used to find your approximation.

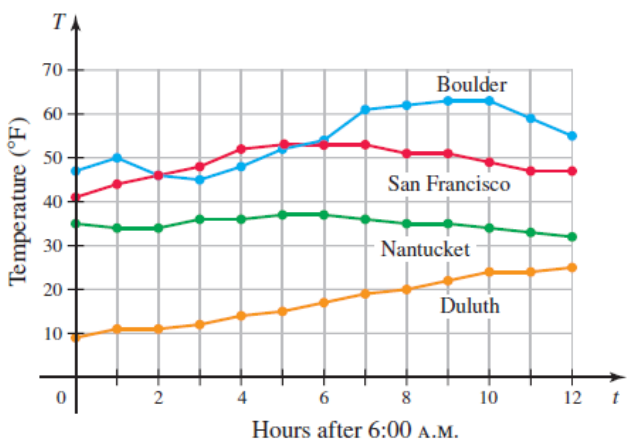
(c) Using the given graph of $y = f(x)$, compute the Simpson's Rule approximation $S(4)$ for $\int_0^8 f(x)\,dx$.Be sure to write out the sum you used to find your approximation.




2. Use Simpson's Rule with $n = 10$ subintervals to approximate $\int_{-2}^{3} e^{-x^2}\,dx$. You may use technology for this calculation.

3. Apply the Midpoint and Trapezoid Rules to $\int_0^{\pi/4} 3\sin(2x)\,dx$ for $n = 4$ subintervals. Find the error, given the exact value of $\frac{3}{2}$.

4. The table shows hourly temperatures $T(t)$ (°F) for Boulder, CO over a 12-hour period. Approximate the average temperature $\bar{T} = \frac{1}{12}\int_0^{12} T(t)dt$ using a numerical method of your choice.




| $t$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B | 47 | 50 | 46 | 45 | 48 | 52 | 54 | 61 | 62 | 63 | 63 | 59 | 55 |
| SF | 41 | 44 | 46 | 48 | 52 | 53 | 53 | 53 | 51 | 51 | 49 | 47 | 47 |
| N | 35 | 34 | 34 | 36 | 36 | 37 | 37 | 36 | 35 | 35 | 34 | 33 | 32 |
| D | 9 | 11 | 11 | 12 | 14 | 15 | 17 | 19 | 20 | 22 | 24 | 24 | 25 |

5. Use Simpson's Rule to approximate $\int_0^{\pi} \sin(6x)\cos(3x)\,dx$.

   (a) Use a trig identity to find the *exact* value of the integral.

   (b) Find a value of $n$ so that your error is less than $10^{-3}$. You may use technology for these calculations.

6. Let $f(x) = \sin(e^x)$.

   (a) Find a Trapezoid Rule approximation to $\int_0^1 \sin(e^x)\,dx$ using $n = 40$ subintervals.You may use technology for this calculation.

   (b) Calculate $f''(x)$.

   (c) Explain why $|f''(x)| \leq 6$ on $[0, 1]$.(Hint: use a graph.)

   (d) Find an upper bound on the absolute error in the estimate im part a.

7. Let $f(x) = \sqrt{\sin x}$.

   (a) Use Simpson's Rule to approximate $\int_1^2 \sqrt{\sin x}\, dx$ with $n = 20$ subintervals.

   (b) Find an upper bound on the absolute error in the estimate found in part a. Use the fact that $f^{(4)}(x) \leq 1$ on $[1, 2]$

8. For what values of $p$ does $\int_1^\infty x^{-p}\, dx$ converge?

9. Evaluate $\int_{-\infty}^{\infty} \frac{dx}{x^2 + 100}$ or state that it diverges.(Hint: the graph of $\arctan(x)$ may be useful.

10. Evaluate $\int_{-\infty}^{-2} \frac{1}{z^2} \sin\left(\frac{\pi}{z}\right) dz$ or state that it diverges.

11. Evaluate $\int_{-\infty}^{\infty} xe^{-x^2}\, dx$ or state that it diverges.

12. Evaluate $\int_{0}^{1} \frac{x^3}{x^4 - 1}\, dx$ or state that it diverges.

13. Evaluate $\displaystyle\int_1^{11} \frac{dx}{(x-3)^{2/3}}$ or state that it diverges.

14. Evaluate $\displaystyle\int_0^1 \ln x \, dx$ or state that it diverges.

15. The region bounded by $f(x) = (x-1)^{-1/4}$ and the $x$-axis on the interval $[1, 2]$ is revolved about the $x$-axis. Find the volume of the solid of revolution.

16. Use a comparison test to determine whether $\displaystyle\int_1^\infty \frac{dx}{x^3+1}$ converges or diverges.

17. Use a comparison test to determine whether $\int_1^\infty \frac{\sin^2 x}{x^2}\,dx$ converges or diverges.

18. Use a comparison test to determine whether $\int_0^1 \frac{dx}{\sqrt{x^{1/3}+x}}$ converges or diverges.